

$$1 \quad (a) \quad m|a \Rightarrow \exists x \in \mathbb{Z}, a = mx \quad (1)$$

$$m|a-b \Rightarrow \exists y \in \mathbb{Z}, a-b = my \quad (2)$$

$$(1) - (2) \Rightarrow b = m(x-y)$$

Since $x-y \in \mathbb{Z}$, thus $m|b$.

It is true that $m|b$.

For a and $a-b$ to have m as the greatest common divisor, then x and y in (1) and (2) have to be relatively prime.

$$(b) \quad \text{For } (i_a, j_a) \neq (i_b, j_b)$$

$$\Rightarrow i_a \neq i_b \text{ or } j_a \neq j_b$$

$$\Rightarrow a \bmod m \neq b \bmod m, \text{ or } a \bmod n \neq b \bmod n$$

$$\Rightarrow a \text{ is not congruent } \overset{b}{\text{ modulo } m}, \text{ or } a \text{ is not congruent } b \text{ modulo } n$$

$$\Rightarrow a-b \text{ is not divisible by } m \text{ or not divisible by } n$$

(c) For two integers congruent modulo 7, they must have the same remainder when divided by 7. There are 7 different remainders (0, 1, 2, ..., 6), so there will be 7 rows in the matrix. There must be at least $\lceil \frac{30}{7} \rceil = 5$ columns in the matrix, i.e., the minimum size of A is 7×5 .

let i be the index of rows, we then have $i = (n \bmod 7) + 1$

$$\forall 1 \leq n \leq 30$$

let j be the index of columns, we have $j = \lceil \frac{n}{7} \rceil, \forall 1 \leq n \leq 30$

$$2. (a) \sum_{i=3}^7 5 = (7-3+1) \times 5 = 25$$

$$\sum_{i=1}^2 \sum_{j=1}^4 (i-j) = \sum_{i=1}^2 ((1-1) + (1-2) + (1-3) + (1-4)) + ((2-2) + (2-3) + (2-4))$$

$$= -9$$

$$\sum_{x \text{ is prime } \wedge x < 7} x^2 = 2^2 + 3^2 + 5^2 = 4 + 9 + 25 = 38$$

(b) Basis step: when $n=7$ l.h.s. $= 11 \times 7 + 17 = 94$
r.h.s. $= 2^7 = 128$

clearly, l.h.s. $\leq$ r.h.s.

Inductive step:

Assume for all $n \geq 7$, that $11n + 17 \leq 2^n$

We'd like to prove that $11(n+1) + 17 \leq 2^{n+1}$

$$\text{l.h.s.} = 11(n+1) + 17$$

$$= (11n + 17) + 11$$

$$\leq 2^n + 11$$

$$< 2^n + 2^7$$

$$\leq 2^n + 2^n$$

$$= 2 \cdot 2^n = 2^{n+1}$$

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(c) Basis step: To prove that 2 people stand in a line, then there is a woman directly in front of a man.

Inductive step: Assume that for $n \geq 2$, n people in a line, there is somewhere in a line a woman directly in front of a man.

Under this assumption, to prove that $(n+1)$ people in a line there is a woman directly in front of a man somewhere in a line.